

Amrinder Singh

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Education

University of Houston, Houston, TX

Bachelor of Science, Computer Science

Expected May 2026

Relevant Coursework: Data Structures & Algorithms, OS, Databases, Software Engineering, Computer Networking, GPU Programming (CUDA), Artificial Intelligence

Experience

Undergraduate Student Researcher, University of Houston · Full-time

Jan 2026 – Present

- Researching improvements to offline search using embeddings and RAG pipelines as part of the IIAB project
- Evaluating system performance on low-end, low-cost Android devices and Raspberry Pi hardware
- Developed FastAPI backend with multi-ZIM Kiwix search, FAISS semantic reranking (99% similarity), and SQLite caching — reducing query time from 6s to 0.10s (98% faster)

Software Engineer, Stealth · Internship

Jan 2026 – Present

- Engineered AR collaboration features for Meta Quest 3, enabling real-time multi-user interaction
- Architected Unity WebRTC pipeline with Azure Web PubSub signaling for low-latency real-time communication
- Built PassthroughCameraAccess-based AR pipeline with Unity and AR Foundation for plane detection and 3D annotation anchoring

Project Experience

IIAB Offline Search Engine, Open-Source Contributor · IIAB Team

January 2026 – Present

- Contributed offline AI search layer to IIAB project, enabling semantic search over 6 ZIM libraries (Wikipedia, DevDocs) on a Raspberry Pi 5 WiFi hotspot serving 20+ concurrent users
- Reduced average query latency by 60% via parallel multi-ZIM scraping (ThreadPoolExecutor) and non-blocking SQLite caching with 7-day TTL, achieving ~95% cache hit rate on repeat queries
- Integrated FAISS semantic reranking (snowflake-arctic-embed) and qwen2.5:0.5b streaming answer box, improving result relevance over keyword-only search

Local Hosted LLM & RAG on Raspberry Pi

Aug 2025 – Dec 2025

- Deployed self-hosted AI dashboard on dual Raspberry Pi 4 edge cluster (4 GB + 1 GB) running Ollama LLM inference and Qdrant vector DB, achieving sub-3s response times on sub-2B parameter models
- Built Web-RAG pipeline (SerpAPI + Cheerio + nomic-embed-text + Qdrant) with SSE streaming, reducing hallucinations and providing up-to-date answers with cited sources unavailable to the base model
- Implemented PDF semantic Q&A (chunk → embed → top-K retrieval → LLM) and WebSocket hardware telemetry dashboard; exposed securely via Cloudflare Tunnel to a GitHub Pages frontend

SpringAI Web Tour Builder

January 2023 – May 2023

- Built full-stack web platform auto-generating interactive onboarding tours using Anthropic Claude API
- Integrated Microsoft OAuth authentication and managed deployment for 50+ demonstration projects
- Created dynamic tour generation system with customizable templates and user analytics dashboard

SiriMind iOS App

May 2021 – Aug 2021

- Developed voice-enabled iOS application integrating OpenAI GPT API for intelligent conversational interface
- Implemented Firebase authentication system supporting 100+ test users with secure data management
- Designed responsive UI using UIKit with speech-to-text and text-to-speech functionality

Technical Skills

Programming Languages: Swift, Java, JavaScript, Python, C++, CUDA

Mobile / XR Development: iOS (UIKit, SwiftUI, Core Data, ARKit), Unity (AR Foundation, Meta XR SDK), Meta Quest 3

Web Technologies: React, Node.js, Express.js, FastAPI, HTML5, CSS3, Django Bootstrap

AI / ML: Ollama, FAISS, Qdrant, ChromaDB, RAG pipelines, WebRTC, embeddings (nomic, snowflake-arctic)

Databases & Cloud: PostgreSQL, MongoDB, Firebase, SQLite, Microsoft Azure, Azure Web PubSub

Tools & APIs: Git/GitHub, Docker, REST APIs, OAuth 2.0, OpenAI GPT API, Anthropic Claude API, Cloudflare Tunnel